

8.2 Example: Converting a Python Script to PHP Code

The following example also comes from our everyday work, but is somewhat more extensive. The starting point is a Python script about 100 lines long. It reads a text file containing database names. The script then analyzes more than 100 similar databases and creates a statistical summary of customer activity. The result is a hard-to-read ASCII table that is sent via email every month.

We want to turn it into a PHP page that can be called at any time. The result is supposed to be displayed as an HTML table. For the code translation, we used ChatGPT 4o; all other work was done in VS Code with GitHub Copilot.

In this example, we'll focus on explaining the prompts. We've refrained from printing the very project-specific code in PHP or Python as it doesn't really contribute to understanding the procedure.

Prompt: I have a Python script that is run monthly to summarize account activity. I want to convert this script to PHP. The result is to be presented in an HTML table instead of ASCII. Can you help please?

[100 lines of Python code]

Certainly! Here's how you can convert the provided Python script to PHP and modify it to present the result in an HTML table.

[170 lines of PHP code]

[more details, explanation, running the script]

The resulting code leaves a good impression at first glance. But we have two fundamental problems:

- First, ChatGPT mixes HTML code with embedded PHP code when outputting the results. This is quite typical of PHP, but we prefer consistent PHP code that performs all output via `echo`. This makes it easier for us to maintain an overview of the code.
- Secondly, the code isn't organized as a function. However, we want to separate the output of statistical data from other subfunctions.

So, we ask for improvements. For this purpose, we first select a code section. The following prompt refers to the selected code. `result` is a variable defined in the previous code that contains the data previously determined from the databases in an array. The highlighted code analyzes this data and turns it into an HTML table.

Prompt: Could you convert this code into a function (with parameter `$result`) using `echo` to output?

Certainly! Here's how you can convert the provided Python script to PHP and modify it to present the result in an HTML table.

[entire code, including the new function `displayAccountActivitySummary...`]

On this basis, we copy the code generated by ChatGPT to VS Code and make various adjustments there. In particular, we use a class for establishing the connection to the MySQL database that is already available for this purpose in our PHP project. In VS Code, we're supported in our work by GitHub Copilot.

Just fifteen minutes after starting work, we're able to carry out an initial test, which is generally satisfactory. For the next fifteen minutes, we optimize the HTML table layout, improve the labeling of the columns, and add a legend to the table. GitHub Copilot surprises us with almost clairvoyant suggestions for the texts, which we adopt with minimal changes.

ChatGPT has adopted the criteria for sorting the results unchanged from the Python script. This basically works, but we want to have a second sort-

ing criterion. However, we have little experience with the PHP function `usort` used by ChatGPT.

Prompt: PHP/`usort`: How do I sort using two criteria instead of one?

We had actually expected a universal answer. Instead, ChatGPT responds with a few lines of code that uses exactly the attribute of the results we had in mind anyway as the second sorting criterion. Without any further changes, the code does what we intended it to do but didn't say. That's a bit uncanny.

8.2.1 Conclusion

The path to a new PHP script presented here is a prime example of how well AI tools can work. We were surprised ourselves at how quickly we reached our destination. Before you object that this is a contrived example, note that we carried out the work outlined here for a project that has nothing to do with this book. We only came up with the idea of presenting the prompts in this chapter at a later stage. So, this very example originates precisely from our real-life work.